Details on the pre-strata
Supporting document `IndigoCarbon_US-1_2024_0068`

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Changes in this supporting document since the last verification

- New strata that have joined the project are included in the summaries in this document

1 Pre-strata

As discussed in the Sample Design section of the Monitoring Plan, the project is partitioned into strata and then random points are selected within each stratum. These strata are called "pre-strata" because they are chosen when the sample points are randomly selected, before observing data at the sample points. To avoid confusion with post-strata (see Supporting Document `IndigoCarbon_US-1_2023_0062` (2023)), we will use the term pre-strata throughout this document.

The project has thirteen pre-strata listed in Table 1. In the database, the pre-strata are referred to by their `stratum_id`, shown in the right-hand column of Table 1. Colloquially, we often refer to the pre-strata by their shorthand name, shown in the left-hand column of Table 1.

Table 1: Names of the pre-strata. The shorthand names are used colloquially by Indigo to refer to the pre-strata, while the pre-stratum IDs are the values stored in the database.

Shorthand name	Pre-stratum ID (used in the database)
A	fall_2019_spring_2020_campaign
B1	fall_2020_outside_texas
B2	fall_2020_texas
C	fall_2020_farmlogs_and_other_stragglers_randomized_late
D	spring_2021
E1	fall_2021
E2	fall_2021_stragglers
E3	fall_2021_stragglers_round_2
E4	spring_2022_stragglers
E5	spring_2022_non_randomized_stragglers
F	spring_2023
G	spring_2024
H	car1459_stratum_h_initial_survey_spring_2025

1.1 Sample design and randomization dates

The dates when pre-strata were randomized (i.e., when the list of sample points were chosen, and—in the case of the two-stage sample design—the list of field selections were made) are listed in Table 2. That table also shows the sample design used in each randomization and the number of points that were chosen to be sampled. (For definitions of those sample designs, please refer to the Sample Design section of the Monitoring Plan.)

The first pre-stratum was Pre-stratum A, which was sampled using a static, national map of random points that was generated once in summer 2019. Pre-stratum A didn't have a "randomization date" because it had ongoing enrollment from September 1, 2019 through August 31, 2020: fields simply joined the pre-stratum when the points that happened to fall within their boundary were soil sampled. Because there was no notion of selecting a field in the sample design for Pre-stratum A, we do not write randomization dates in Table 2.

In fall 2020, the sample design called Two-stage PPSWR/SRSWR (short for "probability proportional to size with replacement" and "simple random sample with replacement") was adopted and has been used on all new pre-strata since then. We aim to make one stratum on an infrequent basis, but as Table 2 shows, some strata were created days apart. To explain one such example, in September 2020, eagerness to begin sampling in Texas before the opportunity to sample closed compelled us to randomize fields in Texas first (forming Pre-stratum B2), before the the boundaries of fields elsewhere were ready to be randomized. The fields in the rest of the country were randomized five days later, on September 8, forming Pre-stratum B1. (Note that in the first verification of this project, Pre-stratum B2 did not appear in the project because those fields' management data were not ready in time; in that first verification, Pre-stratum B1 was called Stratum B.) A small number of fields were randomized a month later, in mid-October 2020, because their boundaries were not ready for randomization in early September 2020; that pre-stratum is known as Pre-stratum C.

A few pre-strata were randomized a second time in order to increase the sample size. Those pre-strata show up multiple times in Table 2. (Because that two-stage sample design selects fields independently with replacement, randomizing those pre-strata a second time is equivalent to randomizing them once but with a larger sample size. It also means that a field could be selected in both randomizations; in that case, we revisit the field to sample the new points selected in it.)

1.2 Locations of fields in the pre-strata

Figure 1 shows the locations of fields' centroids on a map of the USA, with colors and marker shapes denoting their pre-stratum. Note that the pre-strata largely overlap the same regions. Thus, this pre-stratification is essentially stratifying by the time when fields enrolled rather than by location in the country. This choice was driven by practical concerns: fields enrolled slowly over time, and we needed to get started sampling early to be able to achieve a large sample size, so we sampled a random selection of points in each cohort as they enrolled.

1.3 Pre-stratum sizes and sample densities

Table 3 shows the size of each pre-stratum (in terms of fields and acreage) and the density of planned sample points in each pre-stratum. These sample densities are computed for fields that remain in the project; the pre-strata originally contained more fields (and sample points), but some fields dropped out of the program after sampling plans were made and before verification.

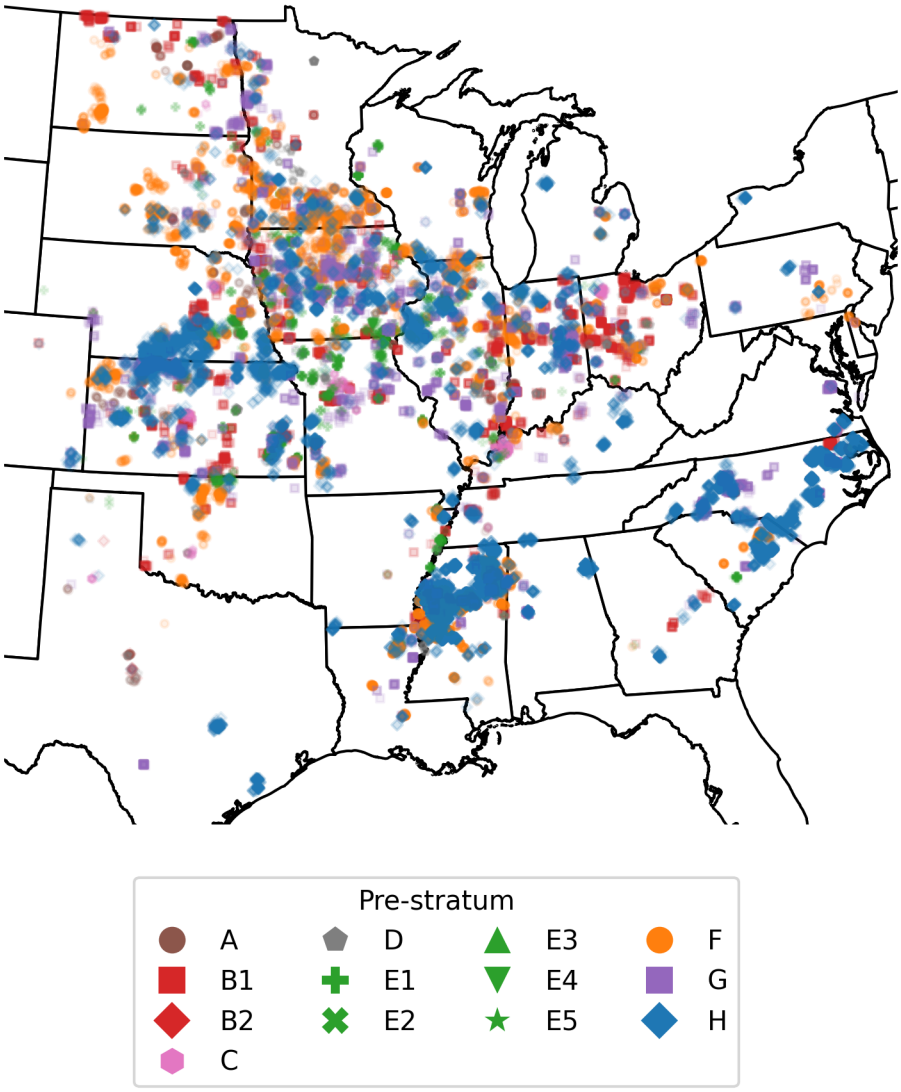


Figure 1: Locations of all fields' centroids and the assignment of fields to pre-strata

Table 2: Date(s), sample design, and the sample size of every randomization on every pre-stratum. Note that pre-stratum A had rolling enrollment between September 1, 2019 through August 31, 2020 because there was no notion of selecting a random subset of fields in stratum A; thus, the randomization date for pre-stratum A is listed as 'N/A' in this table.

Pre-stratum	Date randomized	Sample design	Planned points
A	N/A	Poisson random sample with replacement	9406
B1	Sep 08, 2020	Two-stage PPSWR/SRSWR	3880
B1	Mar 25, 2021	Two-stage PPSWR/SRSWR	7884
B2	Sep 03, 2020	Two-stage PPSWR/SRSWR	24
B2	Mar 25, 2021	Two-stage PPSWR/SRSWR	48
C	Oct 16, 2020	Two-stage PPSWR/SRSWR	156
C	Mar 25, 2021	Two-stage PPSWR/SRSWR	2528
D	Mar 25, 2021	Two-stage PPSWR/SRSWR	2364
E1	Sep 23, 2021	Two-stage PPSWR/SRSWR	5370
E1	Mar 21, 2022	Two-stage PPSWR/SRSWR	1740
E2	Oct 18, 2021	Two-stage PPSWR/SRSWR	12
E3	Nov 01, 2021	Two-stage PPSWR/SRSWR	12
E4	Mar 28, 2022	Two-stage PPSWR/SRSWR	216
E5	May 06, 2022	Two-stage PPSWR/SRSWR	168
F	Mar 10, 2023	Two-stage PPSWR/SRSWR	9250
G	Mar 18, 2024	Two-stage PPSWR/SRSWR	1488
H	Mar 08, 2025	Two-stage PPSWR/SRSWR	3000

1.3.1 Small pre-strata

As the first two columns in Table 3 show, some pre-strata are much smaller than others. The cause of this disparity is that when we make a large pre-stratum, a small number of fields are overlooked or are not quite ready to be included, as mentioned in Section 1.1. Because sampling windows are brief (lasting just a few months in spring and in fall), we randomize as many fields as possible that are ready before a sampling window. In the days after the randomization, we collect these "straggler fields" and randomize them separately. For example, we intended to make one Pre-stratum E, but a few groups of stragglers were discovered in the weeks after creating Pre-stratum E1. When we randomize those "straggler strata" (e.g., Pre-stratum E2), we choose the sample size that makes its sample density as close as possible to that of the stratum for which we intended those straggler fields to belong. For example, Pre-stratum E2 had sample size (in field selections per acre) as close as possible to that of Pre-stratum E1 (and similarly for Pre-strata E2 and E1, E3 and E1, and so on).

After choosing sampling locations, attrition can occur among the randomized fields, and randomness in that attrition results in those sample densities diverging slightly, as can be seen in the "Points/acre" column in Table 3. However, the sample densities remain similar among all pre-strata that were randomized with the two-stage design, with the sparsity of (planned) sample points ranging from 16.18 to 274.24 acres per point, as most are close to the average value shown in the bottom row of Table 3.

Table 3: Density of sample points by pre-stratum

Pre-stratum	Fields	Area (acres)	Points	≥ 3 points	Acres/point	Points/acre
A	671	57,580.9	9,406	✓	6.12	0.163
B1	3,132	228,477.3	11,764	✓	19.42	0.051
B2	8	1,758.8	72	✓	24.43	0.041
C	513	43,431.1	2,684	✓	16.18	0.062
D	817	64,327.2	2,364	✓	27.21	0.037
E1	1,777	154,842.0	7,110	✓	21.78	0.046
E2	3	384.5	12	✓	32.04	0.031
E3	2	205.9	12	✓	17.16	0.058
E4	30	4,423.8	216	✓	20.48	0.049
E5	54	5,583.4	168	✓	33.23	0.030
F	10,617	809,912.6	9,250	✓	87.56	0.011
G	4,645	332,625.8	1,488	✓	223.54	0.004
H	16,278	822,713.0	3,000	✓	274.24	0.004
All	38,547	2,526,266.3	47,546	✓	53.13	0.019

1.3.2 SEP requirement on minimum sample size

The SEP requires that each stratum has at least three points, so that the total and variance can be estimated (*Soil Enrichment Protocol v1.1* 2021, Table 6.2, page 74). The requirement that each stratum contain at least three points is satisfied for all pre-strata, as shown in the "≥ 3 points" column in Table 3.

2 Combining pre-strata into analysis strata

To address the small pre-strata, we combine certain pre-strata together, as shown in Table 4. We call the result of combining one or more strata *analysis strata* because it is a stratum that is used in downstream analysis. We combine strata that had similar sample densities and that were randomized at similar times (see Table 2) and therefore likely have similar characteristics (e.g., the similar distribution across regions can be seen in Fig. 1).

Table 5 shows the result of this combination of pre-strata on the stratum sizes and sample densities. The requirement that each stratum contain at least three points is satisfied for all analysis strata, as shown in the "≥ 3 points" column in Table 5.

References

[1] *Soil Enrichment Protocol v1.1*. May 2021. URL: https://www.climateactionreserve.org/wp-content/uploads/2022/06/Soil-Enrichment-Protocol-V_1.1-final.pdf.

[2] Supporting document `IndigoCarbon_US-1_2023_0062`. *Addressing incomplete soil sample data in Soil Enrichment Protocol projects*. v4.4.1. 2023.

Table 4: Which pre-strata are combined together for analysis purposes

Analysis stratum	Pre-strata that are combined
A	A
B	B1, B2
C	C
D	D
E	E1, E2, E3, E4, E5
F	F
G	G
H	H

Table 5: Density of sample points by analysis stratum

Analysis stratum	Fields	Area (acres)	Points	≥ 3 points	Acres/point	Points/acre
A	671	57,580.9	9,406	✓	6.12	0.163
B	3,140	230,236.1	11,836	✓	19.45	0.051
C	513	43,431.1	2,684	✓	16.18	0.062
D	817	64,327.2	2,364	✓	27.21	0.037
E	1,866	165,439.6	7,518	✓	22.01	0.045
F	10,617	809,912.6	9,250	✓	87.56	0.011
G	4,645	332,625.8	1,488	✓	223.54	0.004
H	16,278	822,713.0	3,000	✓	274.24	0.004
All	38,547	2,526,266.3	47,546	✓	53.13	0.019